Surface-Code Syndrome Utilization for Fault Identification

Can repeated stabilizer syndromes reveal *where* a device is faulty — not just *that* it is?

Jaehun Han[1] *[co-authors / advisor placeholder]*[1] [1][Institution / Group placeholder]

`pjhhan.research@gmail.com` • Niels Bohr Quantum Summer School 2026

1. Motivation

Surface codes already measure syndromes every round — thousands of stabilizer bits are produced purely to detect and decode errors, then discarded.

> ***"This syndrome stream is a free probe of the device. Can we read its own fault structure out of data we already collect for QEC?"***

Long-term vision: a **closed loop** that recycles QEC data into device knowledge, then feeds it back into better decoding.

Repeated syndrome measurements

Device information learning

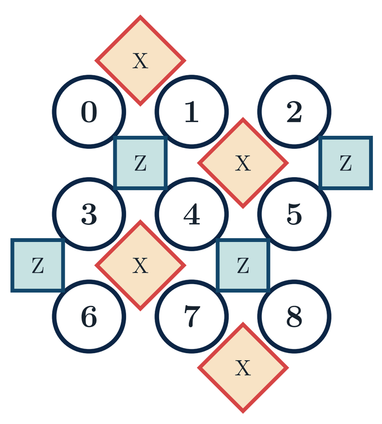
Fault / correlation structure inference

Correlation- / device-aware decoding

feedback

2. Case Study & Problem

Setting. Distance-3 rotated surface code (17 qubits: 9 data + 8 ancilla); each stabilizer-extraction round issues **24 directed CNOTs**. One CNOT is the ***dominant*** fault source (higher fault probability than the rest).



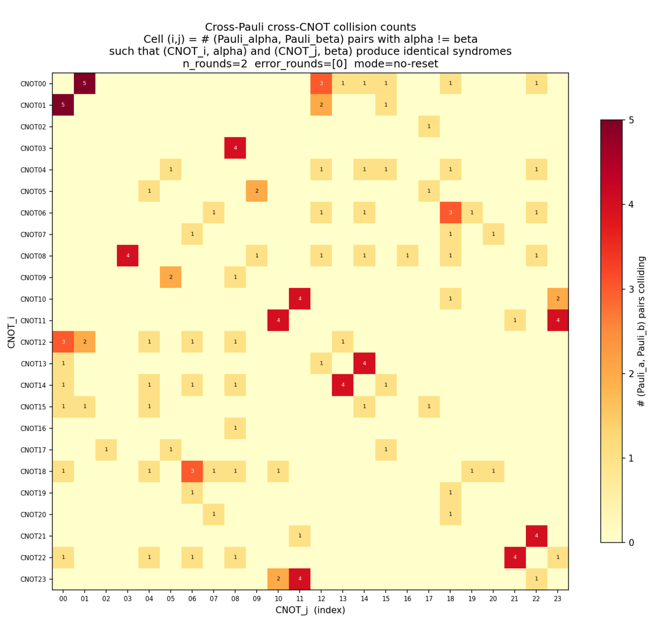
Rotated d=3 layout: 9 data qubits (circles), 8 Z-/X-stabilizer ancillas (squares / diamonds).

The question

From a temporal syndrome / detection-event sequence $S = (s_1, \dots, s_T)$ *alone*, identify the **dominant faulty CNOT** among the 24. **Nuisance factors** (unobserved): which rounds the fault fires, and its Pauli type $(XX, \dots, ZZ)$.

3. Single-Shot Ambiguity

Is one measurement enough? **No.** When the Pauli type is unknown, distinct (CNOT, Pauli) faults can produce *identical* syndromes. Enumerating all 216 single-round cases:



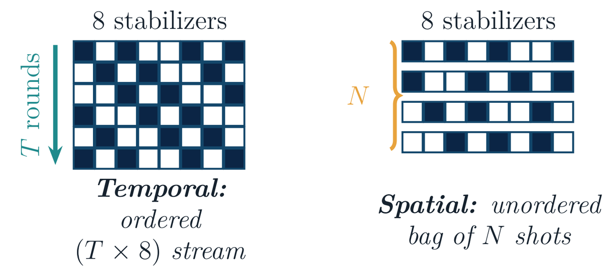
Collision map: cell (i, j) counts Pauli pairs for which faults at CNOT_i and CNOT_j are indistinguishable.

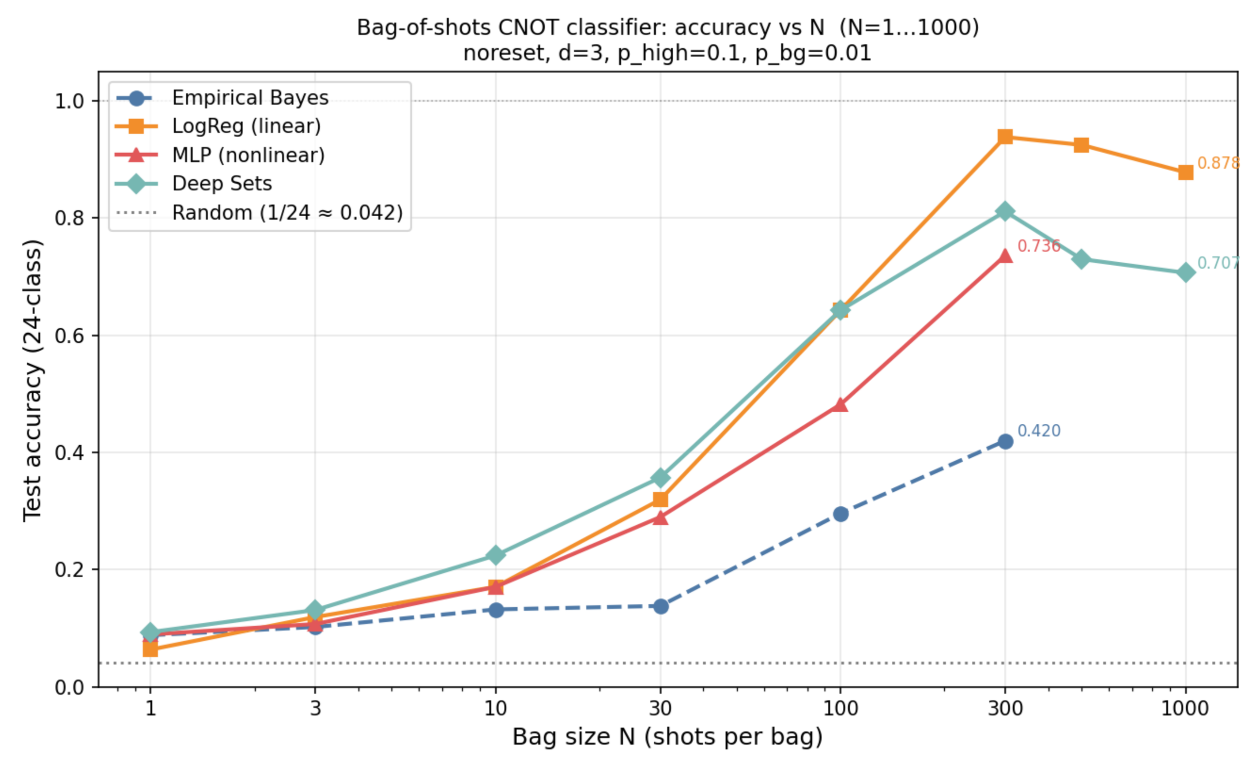
Hard limit

216 cases collapse to 123 signatures with **72** cross-CNOT collisions; a single round with known Pauli identifies only 8/24. **Single-shot 24-class identification is information-theoretically impossible** — invariant under round count and reset mode.

4. More Data Breaks the Ambiguity

The way out is **aggregation**. The same fault is probed either *temporally* (a long syndrome stream) or *spatially* (many repeated single-round shots):





Bag-of-shots CNOT accuracy vs bag size N. At N=1 (single shot) accuracy is near random (1/24); by N=300 it reaches ≈ 0.9 — the ambiguity is broken by data volume, not by a better single shot.

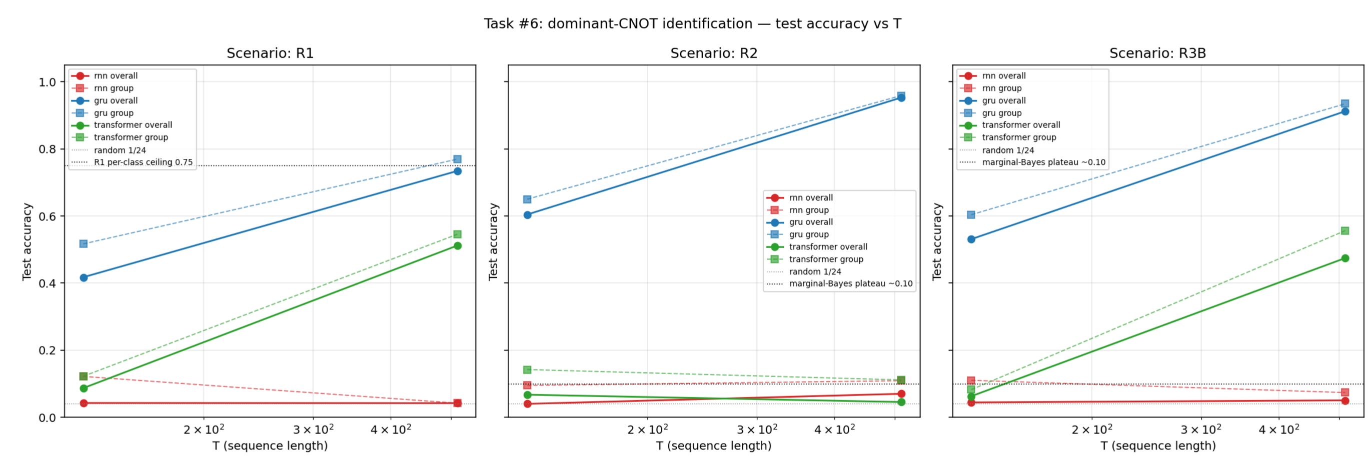
5. Learned Decoders: Models & Performance

Each regime has a matched model class. Best performers so far (**preliminary**, naive d=3):

Temporal: $(T\times 8)$ → **GRU** → dense → softmax (24)

Spatial: bag {shots} → per-shot ϕ → mean pool → MLP ρ → softmax (24)

Deep Sets (spatial) is permutation-invariant over the bag; a linear LogReg on pooled features is a strong baseline.



*Temporal accuracy vs sequence length T (three fault-model scenarios). The **GRU** (blue) climbs with T to ≈95% (R2), breaking the rule-based single-round ceiling (0.75); RNN / transformer lag.*

Headline (preliminary)

Temporal GRU ≈ 95%; spatial LogReg ≈ 94% at N=300. Reported as in-progress, not final performance claims.

6. Status & Outlook

Done. Exact structural map of what syndromes can / cannot reveal about fault location at d=3.

Ongoing. Stim-based *realistic* circuit (parallel schedule, general d); *probabilistic* dominant-fault model.

Next. Close the loop — feed inferred fault / correlation structure into **device-aware decoding**.

Code, data & figures: `Correlated-Noise-Estimation` repository • Contact: `pjhhan.research@gmail.com` • Niels Bohr Quantum Summer School 2026